

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5: Measure network performance using key metrics with simulation tools.
(BL4, BL5)

Assessment Tool Weightage: 30%

Annexure A

Pseudocode Writing Task (Algorithm Design)

Code of Conduct:

(192472188)

I M. Madhulika (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Madhulika

Pseudocode Writing Task (Algorithm Design)

1. In a network simulation using Cisco Packet Tracer, a student records total data received as 8000 bits in 4 seconds and observes that 200 packets were sent while 180 packets were received. Write a pseudocode algorithm to calculate the network throughput in bits per second and the packet loss percentage based on the given values. The algorithm should include validation to ensure that time and total packets sent are not zero to avoid errors. It should display both the throughput and packet loss results clearly. Add logic to classify the network as Efficient if throughput is greater than 1500 bps and packet loss is less than 10%, otherwise classify it as Inefficient. Finally, ensure the algorithm outputs a complete summary of the network performance based on the calculated metrics.
2. An organization has multiple network links connecting its branches. The administrator wants to monitor bandwidth utilization and identify overloaded links. Write a pseudocode algorithm that periodically collects the bandwidth usage of each network link and stores the values in an array or list. Use a loop to continuously monitor all links. Add conditions to detect when bandwidth utilization exceeds 80% and recommend switching traffic to a backup link. Explain how this algorithm supports efficient bandwidth management and prevents network congestion.
3. A company has six branch offices connected through multiple network links. Each link has different bandwidth, delay, and utilization values, which change throughout the day. The network administrator wants to continuously identify the best communication path between the headquarters and each branch based on overall link quality rather than just the shortest distance. Write a pseudocode algorithm that periodically collects the bandwidth, delay, and utilization of every network link and

scores the values in suitable arrays or lists. Use a loop to repeat the monitoring process at regular intervals. Calculate a link quality score for each path using the collected metrics and select the path with the highest score. Include conditions to detect when a link becomes overloaded or fails, automatically selecting an alternate path and generating an alert for the administrator. Explain how your algorithm improves network reliability, routing efficiency, and fault tolerance while adapting to changing network conditions.

RUBRICS FOR CALCULATION

Criteria	Weighting %	Level 4 - Expert	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Performance	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/methods	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Readability Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases, optimized	Handles most cases, reasonable	Handles basic cases only	Incomplete or inefficient

Q1. Network Throughput and Packet Loss

Given:

- Total data received = 8000 bits
- Time = 4 seconds
- Packets sent = 200
- Packets received = 180
- Efficient if throughput > 1500 bps and packet loss < 10%

Pseudocode

```
totalDataReceived = float(input("Enter total data received (bits): "))
time = float(input("Enter time (seconds): "))
packetsSent = int(input("Enter total packets sent: "))
packetsReceived = int(input("Enter total packets received: "))
if time <= 0:
    print("Error: Time must be greater than zero.")
elif packetsSent <= 0:
    print("Error: Packets sent must be greater than zero.")
else:
    throughput = totalDataReceived / time
    packetLoss = ((packetsSent - packetsReceived) / packetsSent) * 100
    print("\n----- NETWORK PERFORMANCE -----")
    print("Throughput :", round(throughput, 2), "bps")
    print("Packet Loss :", round(packetLoss, 2), "%")
    if throughput > 1500 and packetLoss < 10:
        print("Classification: EFFICIENT")
    else:
        print("Classification: INEFFICIENT")
    print("\n----- PERFORMANCE SUMMARY -----")
    print("Data Received      :", totalDataReceived, "bits")
    print("Time                :", time, "seconds")
    print("Packets Sent        :", packetsSent)
    print("Packets Received    :", packetsReceived)
    print("Throughput          :", round(throughput, 2), "bps")
    print("Packet Loss         :", round(packetLoss, 2), "%")
```

INPUT:

```
Enter total data received (bits): 8000
Enter time (seconds): 2
Enter total packets sent: 200
Enter total packets received: 180
```


OUTPUT:

```
----- NETWORK PERFORMANCE -----  
Throughput   : 4000.0 bps  
Packet Loss  : 10.0 %  
Classification: INEFFICIENT
```

```
----- PERFORMANCE SUMMARY -----  
Data Received   : 8000.0 bits  
Time            : 2.0 seconds  
Packets Sent    : 200  
Packets Received : 180  
Throughput      : 4000.0 bps  
Packet Loss     : 10.0 %  
|
```

Calculation

Throughput:

$$8000 / 4 = 2000 \text{ bps}$$

Packet Loss:

$$((200 - 180) / 200) \times 100 = 10\%$$

Since throughput is 2000 bps > 1500 bps, but packet loss is 10% and not less than 10%, the network is:

Network Classification: INEFFICIENT

Q2. Bandwidth Utilization Monitoring

The algorithm must periodically collect bandwidth usage, store it in an array/list, continuously monitor links, detect utilization above 80%, and recommend switching to a backup link

Pseudocode

```
File Edit Format View Options Window Help
n = int(input("Enter number of network links: "))
if n <= 0 or n > 100:
    print("Invalid number of links.")
else:
    bandwidth = []
    print("\n----- BANDWIDTH MONITORING -----")
    for i in range(n):
        value = float(input(
            f"Enter bandwidth utilization of Link {i + 1} (%): "
        ))
        if value < 0 or value > 100:
            print("Invalid utilization value.")
            break
        bandwidth.append(value)
        print(f"Link {i + 1} Utilization: {value:.2f}%")
        if value > 80:
            print(f"WARNING: Link {i + 1} is overloaded.")
            print("Recommendation: Switch traffic to backup link.")
        else:
            print(f"Link {i + 1} is operating normally.")
        print()
    else:
        print("Bandwidth monitoring completed.")
```


OUTPUT:

```
-----  
Enter number of network links: 4  
-----  
BANDWIDTH MONITORING  
Enter bandwidth utilization of Link 1 (%): 60  
Link 1 Utilization: 60.00%  
Link 1 is operating normally.  
  
Enter bandwidth utilization of Link 2 (%): 85  
Link 2 Utilization: 85.00%  
WARNING: Link 2 is overloaded.  
Recommendation: Switch traffic to backup link.  
  
Enter bandwidth utilization of Link 3 (%): 70  
Link 3 Utilization: 70.00%  
Link 3 is operating normally.  
  
Enter bandwidth utilization of Link 4 (%): 95  
Link 4 Utilization: 95.00%  
WARNING: Link 4 is overloaded.  
Recommendation: Switch traffic to backup link.  
  
Bandwidth monitoring completed.
```

Explanation

1. The administrator enters the number of network links.
2. Bandwidth utilization is collected for every link.
3. The values are stored in an array.
4. A FOR loop checks every network link.
5. If utilization exceeds 80%, the link is considered overloaded.
6. The algorithm recommends transferring traffic to a backup link.
7. The WHILE loop continuously monitors the network.
8. Periodic monitoring helps detect congestion early and improves bandwidth management.